

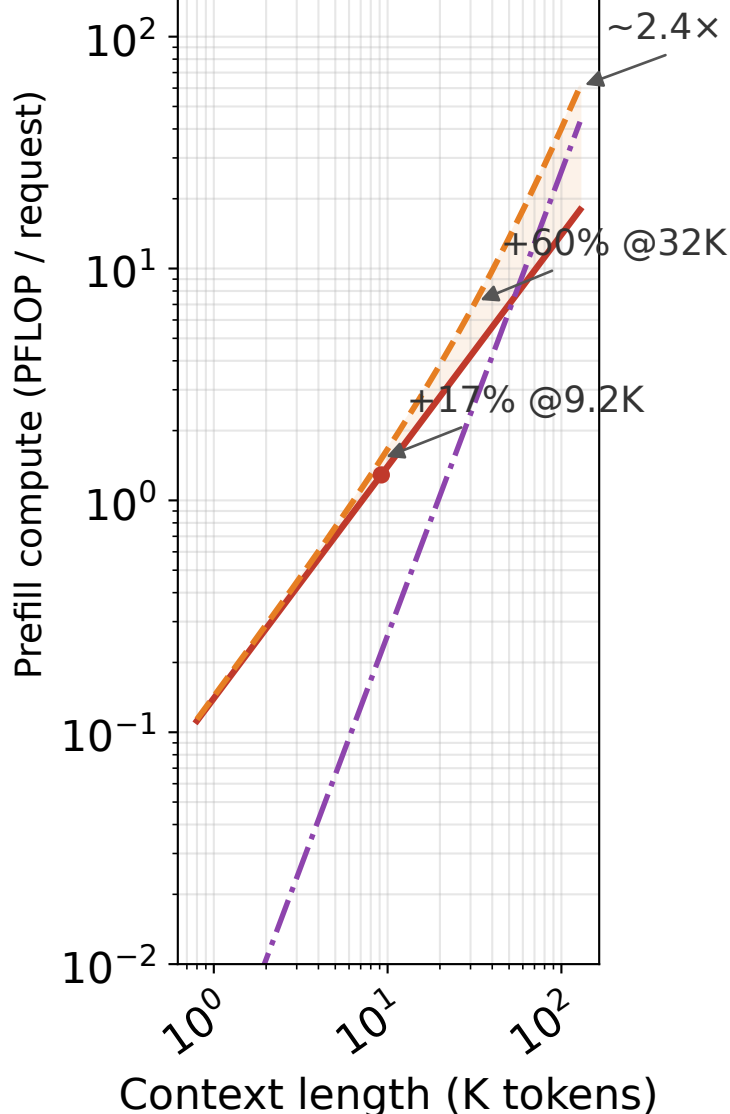


LEFT: PER REQUEST | RIGHT: PER GENERATED TOKEN

DO NOT COMPARE Y-AXIS MAGNITUDES DIRECTLY — different units, different baselines

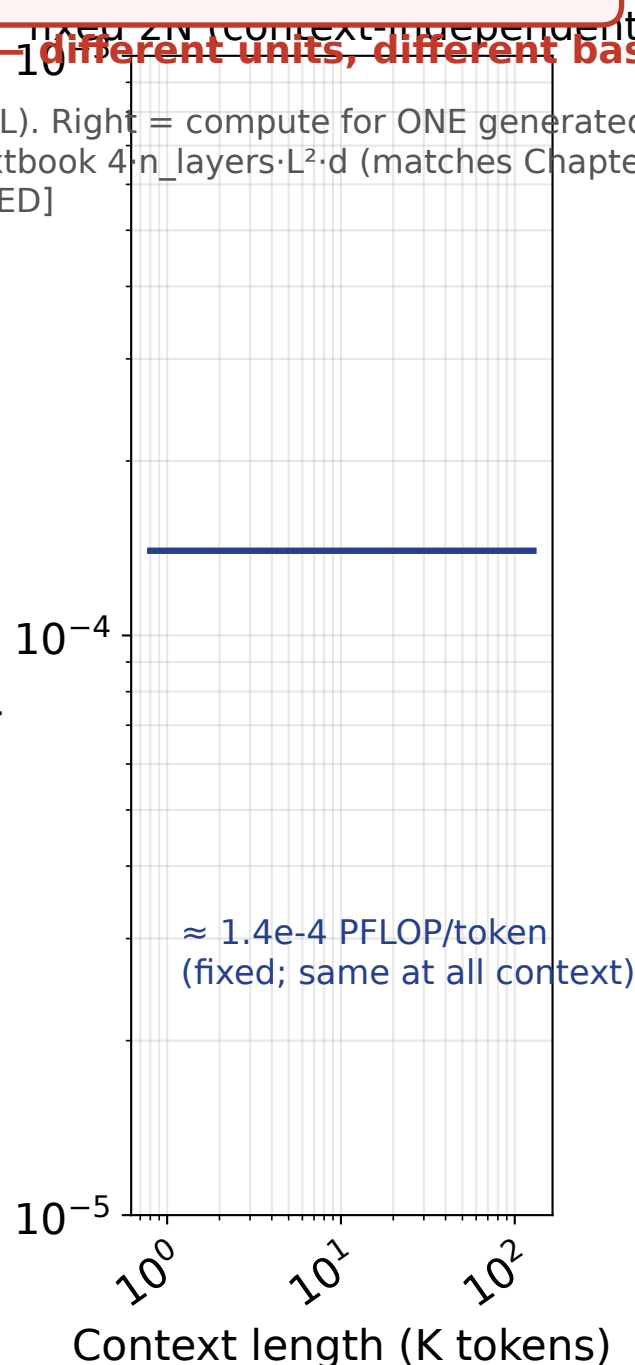
Left = total prefill compute for ONE request (grows with context L). Right = compute for ONE generated token (fixed 2N, context-independent). The quadratic term is the textbook $4 \cdot n_{\text{layers}} \cdot L^2 \cdot d$ (matches Chapter 8).

[ILLUSTRATIVE][DERIVED]



- linear term ($2 \cdot N \cdot L$)
- · - quadratic attention term ($4 \cdot n_{\text{layers}} \cdot L^2 \cdot d$)
- - - total = linear + quadratic

Decode compute (PFLOP / token)



$\approx 1.4 \times 10^{-4}$ PFLOP/token
(fixed; same at all context)